

1. Data Collection
2. Data Cleaning and Preprocessing
3. Feature Engineering using Lag Variables
4. Model Training (ML algorithms)
5. Evaluation (MSE, MAE, SMAPE, R^2)
6. Visualization and Reporting

Implementation Details:

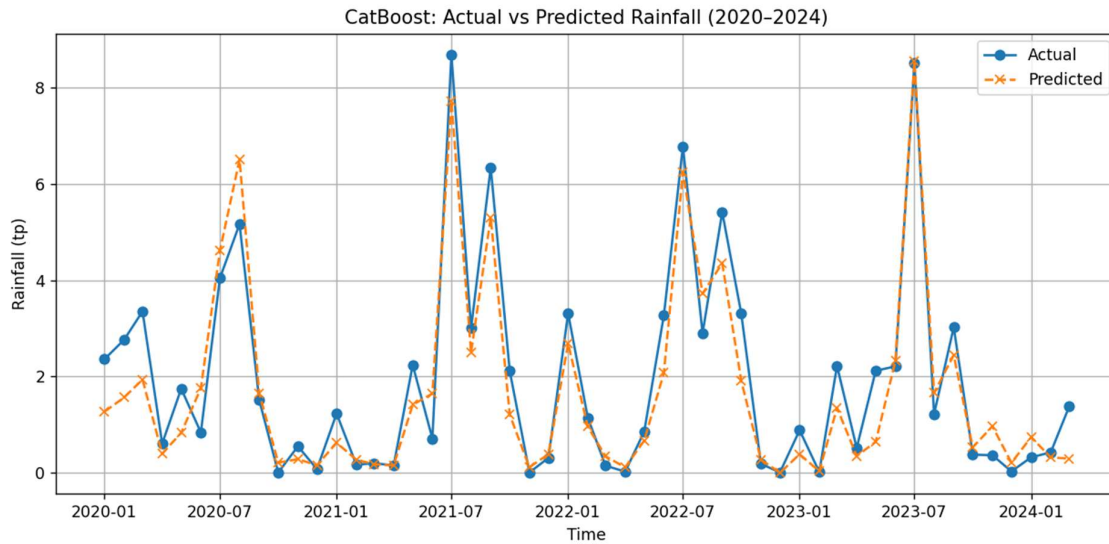
- Lag features were created to capture temporal dependencies (e.g., `tp_lag1`, `tp_lag3`).
- Correlation analysis was used to identify impactful variables.
- Several ML models were evaluated and tuned, including SVR, Random Forest, LightGBM, XGBoost, and CatBoost.
- Best accuracy was achieved using **CatBoost** with an R^2 of 0.8938.
- Evaluation was done using MSE, MAE, SMAPE, and custom SMAPE-based accuracy.

Sample Outputs / Screenshots:

```
cat_boost.py > ...
1  import pandas as pd
2  import numpy as np
3  from catboost import CatBoostRegressor
4  from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
5  import matplotlib.pyplot as plt
6
7  # Step 1: Load the dataset
8  df = pd.read_csv(r"D:\Internship\IMD project Rainfall prediction\delhi_data_era5_all (1).csv")
9
10 # Step 2: Parse the date column and sort
11 df['time'] = pd.to_datetime(df['time'], format='%d-%m-%Y')
12 df = df.sort_values('time')
13
14 # Step 3: Create quarterly lag features (lags of 1, 4, 7, and 10 days)
15 lag_days = [1,4,7,10]
16 for lag in lag_days:
17     df[f'tp_lag{lag}'] = df['tp'].shift(lag)
18
19 # Step 4: Drop rows with missing values
20 df_model = df.dropna().copy()
21
22 # Step 5: Define features and target
23 features = [f'tp_lag{lag}' for lag in lag_days] + [
24     col for col in df.columns if col not in ['time', 'tp'] and not col.startswith('tp_lag')
25 ]
26 X = df_model[features]

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\Internship\IMD project Rainfall prediction\AI & ML model>
PS D:\Internship\IMD project Rainfall prediction\AI & ML model> python cat_boost.py
CatBoost Evaluation:
Mean Squared Error (MSE): 0.4958
Mean Absolute Error (MAE): 0.5417
R2 Score: 0.8938
SMAPE: 57.62%
Accuracy (SMAPE-based): 42.38%
```



Challenges Faced and Solutions:

- **Challenge:** High error variability with certain lag combinations.
Solution: Experimented with different lag strategies (short, medium, quarterly, alternate months).
- **Challenge:** Low correlation of some features.
Solution: Performed correlation-based feature selection.
- **Challenge:** SMAPE accuracy not consistent across models.
Solution: Prioritized R^2 and adjusted feature sets accordingly.

Contribution to the Organization:

- Developed a flexible and reusable forecasting framework.
- Evaluated and benchmarked multiple ML models for operational feasibility.
- Shared insights on variable selection and model interpretability.
- Improved understanding of rainfall predictability patterns for Delhi.

Future Scope of the Project:

- Integrating **deep learning** models like LSTM for sequential forecasting.
- Expanding prediction to **multi-region** or **pan-India** rainfall.
- Deploying the model as a **web-based forecasting tool** for real-time use.
- Using real-time sensor or satellite data to improve accuracy.